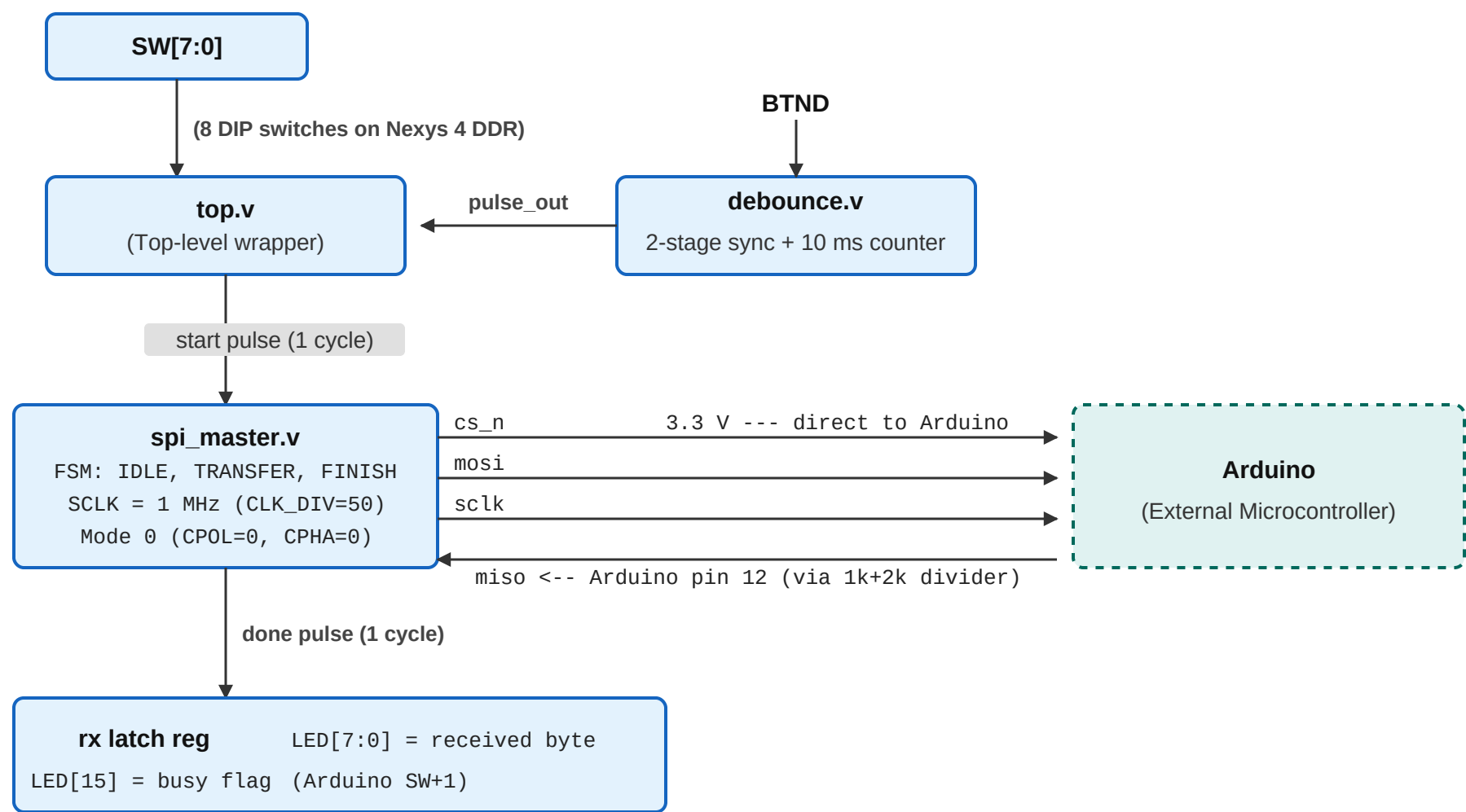


3.1 System Block Diagram



PMOD JA Connections:
JA1(C17)=CS_N JA2(D18)=MOSI JA3(E18)=MISO JA4(G17)=SCLK

Figure 3.1 — System architecture: data flow from switches to FPGA SPI master to Arduino and back to LEDs

3.3 SPI FSM Diagram

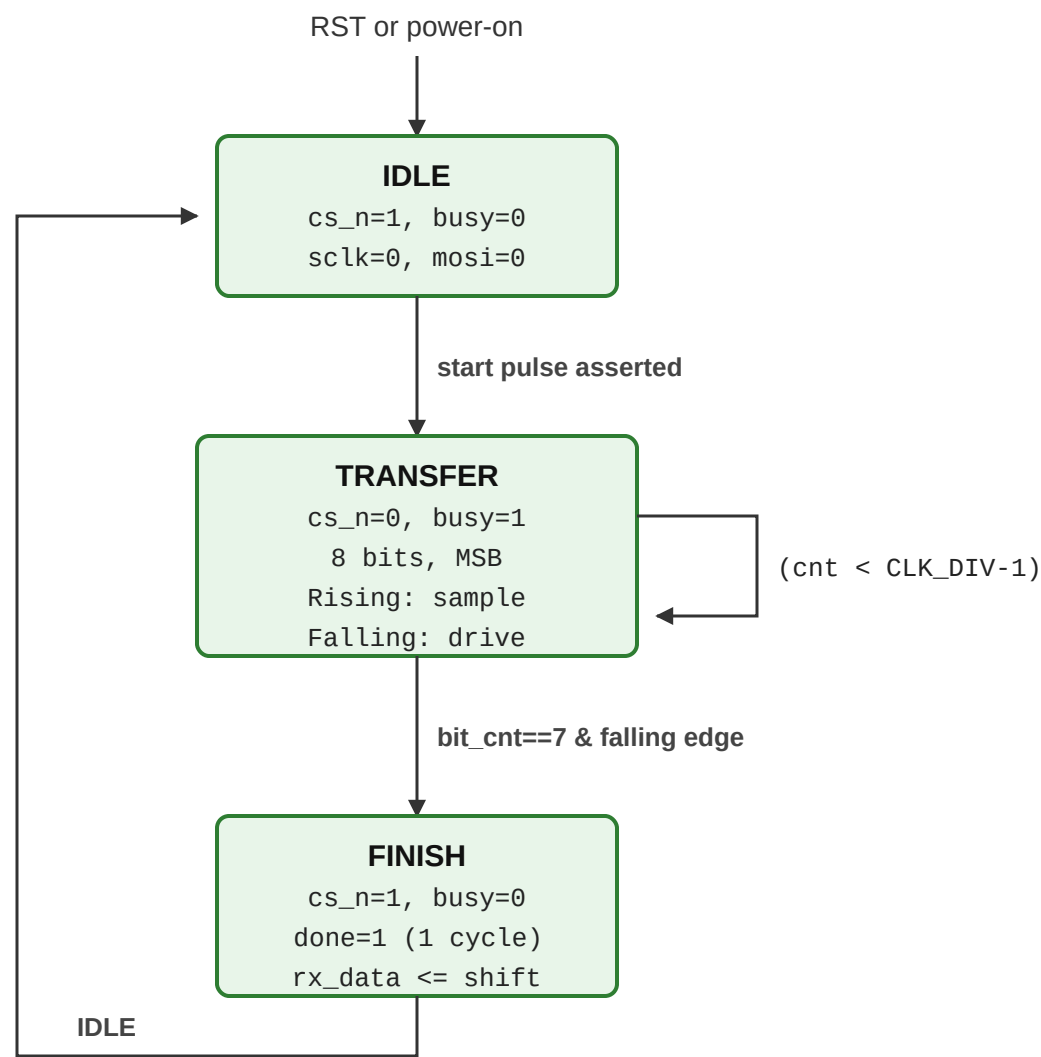


Figure 3.2 — Three-state FSM: IDLE waits for start, TRANSFER shifts 8 bits, FINISH pulses done and returns to IDLE